Calculus I Lesson 21B

1) $y = f(x) = \cos^2 x$

Hint: $\cos^2 x = (\cos x)^2$

Let $u = \cos x$

a) $du/dx =$ ___?___

b) Rewrite $y = (\cos x)^2$ in terms of u: $y =$ ___?___

c) $dy/du =$ ___?___

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} =$ ___?___

2) $y = f(x) = \sin^3 x$

Hint: $\sin^3 x = (\sin x)^3$

Let $u = \sin x$

a) $du/dx = \underline{\quad ? \quad}$

b) Rewrite $y = (\sin x)^3$ in terms of u: $\quad y = \underline{\quad ? \quad}$

c) $dy/du = \underline{\quad ? \quad}$

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

3) $y = f(x) = \tan^2 x$

Hint: $\tan^2 x = (\tan x)^2$

Let $u = \tan x$

a) $du/dx = \underline{\quad ? \quad}$

b) Rewrite $y = (\tan x)^2$ in terms of u: $\quad y = \underline{\quad ? \quad}$

c) $dy/du = \underline{\quad ? \quad}$

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

4) $y = f(x) = \csc^2 x$

Hint: $\csc^2 x = (\csc x)^2$

Let $u = \csc x$

a) $du/dx = \underline{\quad ? \quad}$

b) Rewrite $y = (\csc x)^2$ in terms of u: $\quad y = \underline{\quad ? \quad}$

c) $dy/du = \underline{\quad ? \quad}$

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} = \underline{\quad ? \quad}$

5) $y = f(x) = \tan(4x - \pi)$

Let $u = 4x - \pi$

a) $du/dx =$ ___?___

b) Rewrite $y = \tan(4x - \pi)$ in terms of u: $y =$ ___?___

c) $dy/du =$ ___?___

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} =$ ___?___

6) $y = f(x) = \sin\left(4x^2 + 3\right)$

Let $u = 4x^2 + 3$

a) $du / dx =$ __?__

b) Rewrite $y = \sin\left(4x^2 + 3\right)$ in terms of u : $y =$ __?__

c) $dy / du =$ __?__

d) Rewrite dy / du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} =$ __?__

7) $y = f(x) = \cot\left(5x^{1/2} - 4\pi\right)$

Let $u = 5x^{1/2} - 4\pi$

a) $du/dx =$ ___?___

b) Rewrite $y = \cot\left(5x^{1/2} - 4\pi\right)$ in terms of u: $y =$ ___?___

c) $dy/du =$ ___?___

d) Rewrite dy/du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} =$ ___?___

8) $y = f(x) = \cos(4x)^2$

Note: $\cos^2(4x) = \cos(4x) \cdot \cos(4x)$; and $\cos(4x)^2 = \cos(4x \cdot 4x) = \cos\left(16x^2\right)$

Let $u = 16x^2$

a) $du / dx =$ ___?___

b) Rewrite $y = \cos\left(16x^2\right)$ in terms of u : $y =$ ___?___

c) $dy / du =$ ___?___

d) Rewrite dy / du in terms of x.

e) $y' = f'(x) = \dfrac{dy}{du} \cdot \dfrac{du}{dx} =$ ___?___